Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

COAL PROJECT REPORT

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Semester: 4(A)

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Project Title:

Smart Home Automation System using Arduino Uno and RS232 Serial Communication

1. Introduction:

This project demonstrates a Smart Home Automation System using Arduino Uno and EMU8086 Assembly simulation, where two devices (represented by LEDs) are controlled via serial input (RS232). It simulates basic home control features like turning devices ON/OFF and mimicking automation.

2. Objectives:

- To control home appliances (2 devices) using Arduino and Assembly simulation.
- To demonstrate serial communication via RS232.
- To simulate a smart environment with ON/OFF feedback.
- To implement automation using both hardware and emulator software.

3. Components Required:

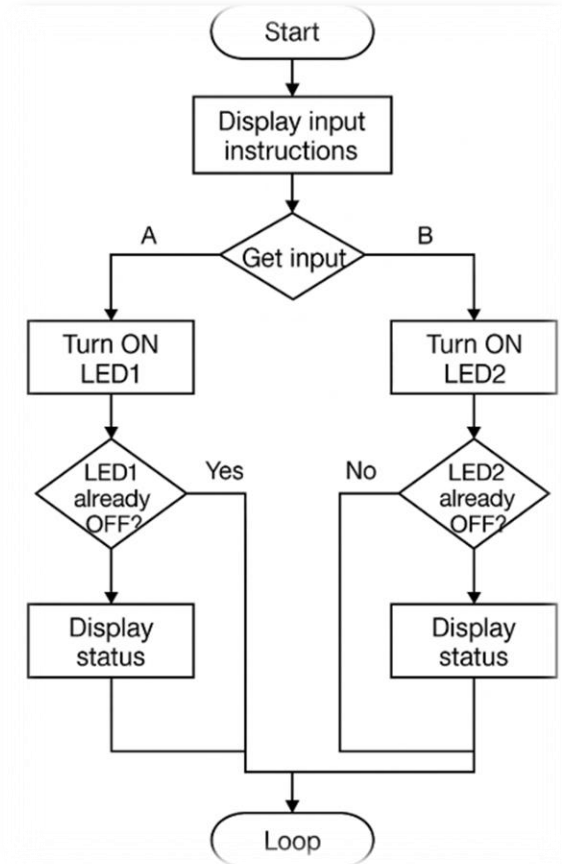
Component	Quantity
Arduino Uno	1
Breadboard	1
LEDs	2
Resistors (220–330Ω)	2
USB to RS232 Converter	1
Jumper Wires	As required
Arduino USB Cable	1

4. Flowchart:

The flowchart describes:

- Waiting for input (A/B to turn ON, a/b to turn OFF).
- Checking the current status.
- Giving proper feedback (e.g., "LED1 already OFF").
- Looping back for the next input.

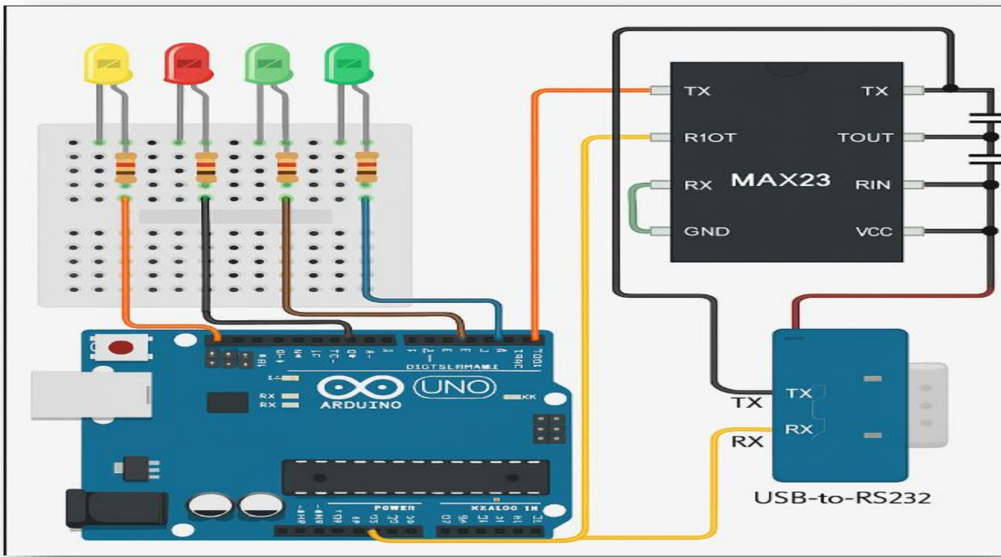
Repeat the Process Continuously



5. Circuit Diagram:

The circuit shows:

- **LED1** connected to **pin 2** of Arduino with a 220Ω resistor.
- **LED2** connected to **pin 3** of Arduino with a 220Ω resistor.
- **MAX232** IC between Arduino TX/RX and RS232 connector.
- **Capacitors** connected to MAX232 for voltage level shifting.



6. Working Principle:

- Arduino reads serial input (via USB or RS232).
- When 'A' is pressed, LED1 turns ON.
- When 'a' is pressed, LED1 turns OFF.
- When 'B' is pressed, LED2 turns ON.
- When 'b' is pressed, LED2 turns OFF.
- If a device is already in the requested state, it displays a message like: "LED1 is already OFF".

7. Complete Hardware Connections (All Components):

1. **LED1**:
   - Positive leg to **pin 2** of Arduino via a 220Ω resistor
   - Negative leg to **GND**
2. **LED2**:
   - Positive leg to **pin 3** of Arduino via a 220Ω resistor
   - Negative leg to **GND**
3. **USB to RS232**:
   - Connects PC to RS232 port.
4. **Arduino USB Cable**:
   - Connects Arduino to PC for programming or serial monitoring.

8. Arduino Code:

CODE:

```
// Pin Definitions
const int led1 = 8;
const int led2 = 9;
void setup() {
  Serial.begin(9600);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
  Serial.println("Press A or a to toggle LED1");
  Serial.println("Press B or b to toggle LED2");
}
void loop() {
  if (Serial.available()) {
    char command = Serial.read();
    switch (command) {
      case 'A':
      case 'a':
        digitalWrite(led1, !digitalRead(led1)); // Toggle LED1
        Serial.print("LED1 is now ");
        Serial.println(digitalRead(led1) ? "ON" : "OFF");
        break;
      case 'B':
```

```
      case 'b':
        digitalWrite(led2, !digitalRead(led2)); // Toggle LED2
        Serial.print("LED2 is now ");
        Serial.println(digitalRead(led2) ? "ON" : "OFF");
        break;
      default:
        Serial.println("Invalid input. Use A/a for LED1, B/b for LED2.");
    }
  }
}
```

9. Assembly Code (8086 EMU8086 Style – Simulation):

CODE:

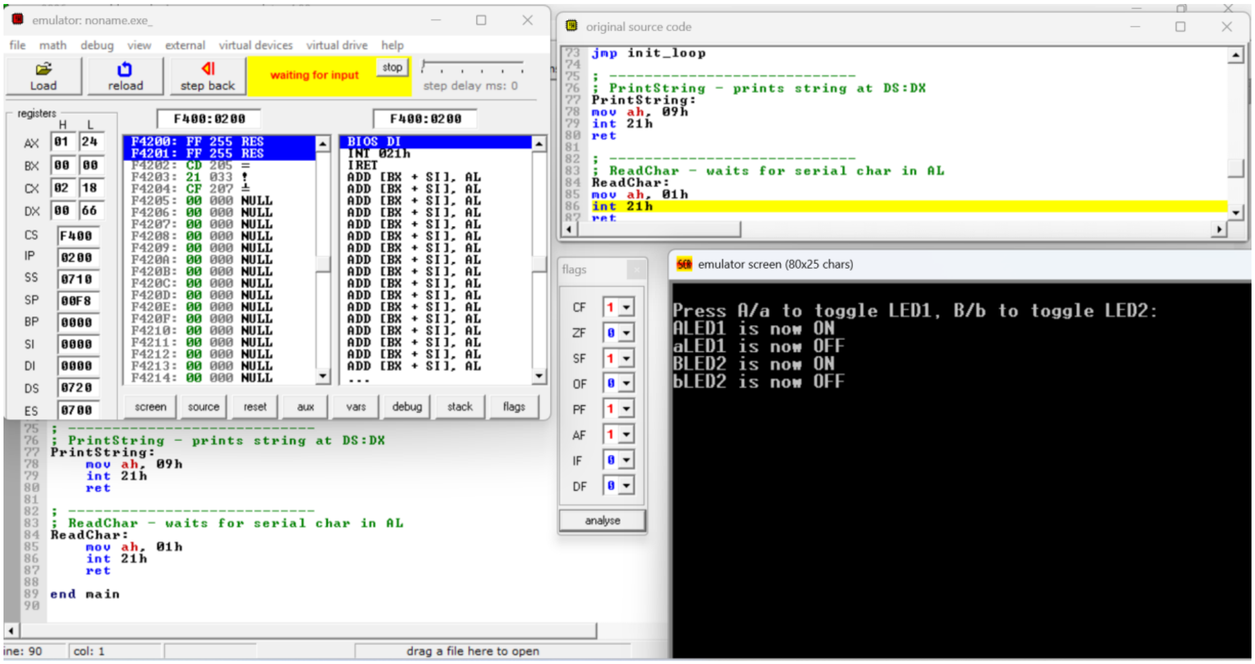
```
.model small
.stack 100h
.data
msg_menu db 13,10,'Press A/a to toggle LED1, B/b to toggle LED2:',13,10,'$'
msg_on1 db 'LED1 is now ON',13,10,'$'
msg_off1 db 'LED1 is now OFF',13,10,'$'
msg_on2 db 'LED2 is now ON',13,10,'$'
msg_off2 db 'LED2 is now OFF',13,10,'$'
msg_invalid db 'Invalid input. Use A/a for LED1, B/b for LED2.',13,10,'$'
led1_state db 0
led2_state db 0
```

```
.code
main:
    mov ax, @data
    mov ds, ax
    ; Display menu once
    mov dx, offset msg_menu
    call PrintString
init_loop:
    call ReadChar
    ; Check input
    cmp al, 'A'
    je toggle_led1
    cmp al, 'a'
    je toggle_led1
    cmp al, 'B'
    je toggle_led2
    cmp al, 'b'
    je toggle_led2
    ; Invalid input
    mov dx, offset msg_invalid
    call PrintString
    jmp init_loop
toggle_led1:
```

```
    ; Toggle LED1 state
    mov al, led1_state
    xor al, 1
    mov led1_state, al
    cmp al, 1
    je show_on1
    mov dx, offset msg_off1
    call PrintString
    jmp init_loop
show_on1:
    mov dx, offset msg_on1
    call PrintString
    jmp init_loop
toggle_led2:
    ; Toggle LED2 state
    mov al, led2_state
    xor al, 1
    mov led2_state, al
    cmp al, 1
    je show_on2
    mov dx, offset msg_off2
    call PrintString
    jmp init_loop
```

```
show_on2:
    mov dx, offset msg_on2
    call PrintString
    jmp init_loop
; ----------------------------
; PrintString - prints string at DS:DX
PrintString:
    mov ah, 09h
    int 21h
    ret
; ----------------------------
; ReadChar - waits for serial char in AL
ReadChar:
    mov ah, 01h
    int 21h
    ret
end main
```

OUTPUT:



10. Implementation & Testing:

Steps:

- Assembled the circuit on the breadboard with LEDs and resistors.
- Connected RS232 converter to Arduino's RX pin.
- Uploaded code via Arduino IDE.
- Opened serial terminal and typed A–B/a–b.

Results:

- Correct LED turns ON with uppercase command.
- LED turns OFF with lowercase.
- Instant response observed.
- RS232 interface worked reliably at 9600 baud.

11. Applications:

- Smart lighting systems
- Basic home/office automation
- Prototyping for IoT applications

12. Future Enhancements:

- Add sensor-based automation (temperature, motion).
- Control via Bluetooth or mobile app.
- Add voice control or integrate with Alexa/Google Assistant.

13. Conclusion:

This project successfully demonstrates the basic working of a smart home automation system using Arduino Uno and RS232 communication. It is scalable and can be upgraded with sensors and wireless modules. It enhances our understanding of microcontroller communication and device control.